

6 (a) Temperature in Celsius = $77.30 - 273.15$
 $= -195.85\text{ }^{\circ}\text{C}$

Note: Some may failed to consider the number of decimal places required for the correct answer. The temperature, in kelvin, was to be correct up to two places of decimals.

(b) This means that the temperature on such a scale is not dependent on the thermometric property of any particular substance.

(c) (i) The *internal energy* of a gas is a **function of the state of the gas**, and is the sum of the random kinetic energy due to the microscopic random motion of its atoms/molecules, and potential energy due to intermolecular forces of attraction.

(ii) An *ideal gas* is a hypothetical gas that obeys the equation of state at all pressure, volume and temperature of the gas.

Equation of state : $pV=nRT$

where p = pressure of the ideal gas

V = volume of the ideal gas

n = number of moles of the ideal gas

R = molar gas constant

T = temperature of the ideal gas

(d) (i) Apply the equation of state of an ideal gas,

$$pV = nRT$$

$$n = \frac{pV}{RT}$$

$$\begin{aligned}\Delta n &= \frac{\Delta p V}{RT} \\ &= \frac{(3.23 \times 10^5 - 2.62 \times 10^5)(0.0120)}{(8.31)(273.15 + 25)} \\ &= 0.296 \text{ mol}\end{aligned}$$

- (ii) To supply four tyres, the amount of gas in moles transferred from the supply to the tyres is

$$\Delta n = 4(0.296) = 1.18 \text{ mol}$$

The fall in the supply's pressure is

$$\begin{aligned}\Delta p &= \frac{\Delta n RT}{V} \\ &= \frac{(1.18)(8.31)(273.15 + 25)}{(0.0108)} \\ &= 2.71 \times 10^5 \text{ Pa}\end{aligned}$$

The supply's pressure will fall to $8.72 \times 10^5 - 2.71 \times 10^5 = 6.01 \times 10^5 \text{ Pa}$

This is still above $3.23 \times 10^5 \text{ Pa}$.

Note: Common errors include failing to allow for the fact that the pressure must not fall below $3.23 \times 10^5 \text{ Pa}$. Instead, they assumed that all of the air in the tank could be used.

- (e) (i) For an ideal gas, the internal energy is its random kinetic energy alone (since potential energy is zero).

$$\begin{aligned}U_{\text{one molecule}} &= \frac{3}{2} kT \\ &= \frac{3}{2} (1.38 \times 10^{-23})(273.15 + 25) \\ &= 6.17 \times 10^{-21} \text{ J}\end{aligned}$$

- (ii) One mole = 6.02×10^{23} molecules.

$$\begin{aligned}\text{Internal energy, } U_{\text{one mole}} &= (6.02 \times 10^{23})(6.17 \times 10^{-21}) \\ &= 3.71 \times 10^3 \text{ J}\end{aligned}$$

- (iii) Increase in internal energy, $\Delta U = (0.296)(3.71 \times 10^3)$
 $= 1.10 \times 10^3 \text{ J}$

- (f) (i) The **increase** in the internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system.

- (ii) Since there is no heat supplied to or removed from the system, $Q = 0$

$$\Delta U = Q + W$$

$$1.10 \times 10^3 = 0 + W$$

Work done **on** the system is $1.10 \times 10^3 \text{ J}$